Notes version: 12 Nov 19:14 IST (if this time is old, reload the page)

GEOGRAPHY · Atmosphere: structure and composition; heat and pressure

UNDERSTAND IT FIRST

What is it? The **atmosphere** is the blanket of air around the Earth, held by **gravity**. It gives us **oxygen to breathe**, **blocks harmful ultraviolet rays**, keeps the planet **warm enough for life** (greenhouse effect) and is where all **weather** happens. The Sun **heats** the Earth unevenly, and differences in heat create differences in **air pressure**, which make **winds** blow (Day 29). Learn it as **gases → layers → how the Earth is heated → heat budget → temperature changes with height → pressure and pressure belts**.

1. Composition

- **Dry air by volume: nitrogen ~78%**, **oxygen ~21%**, **argon ~0.93%** (the most common **inert** gas), **carbon dioxide ~0.04%** (over **420 ppm** and rising); traces of neon, helium, ozone, hydrogen. Most of the air (about **99%**) lies within about **32 km** of the surface.
- **Water vapour** varies from almost **0 to 4%** (most in hot, humid tropics); it absorbs heat and forms **clouds and rain**. **Dust particles** act as **hygroscopic nuclei** around which water condenses. **Ozone** (mainly at **15–35 km**) absorbs the Sun's **ultraviolet** rays. **CO_2 and water vapour** let sunlight in but **trap outgoing heat** (**greenhouse effect**).

2. Layers of the atmosphere

Layers of the atmosphere and temperature

Layer	Features
Troposphere	Lowest layer: about **8 km at the poles** and **18 km at the equator** (thicker where strong heating pushes air up). Holds most of the air, **water vapour and dust**, so **all weather** (clouds, rain, storms) happens here. Temperature **falls** with height at about **6.5°C per km** (**normal lapse rate**). Its top is the **tropopause**.
Stratosphere	Up to about **50 km**. Contains the **ozone layer**; because ozone absorbs UV, temperature **rises** with height. Almost **no clouds or weather** and little turbulence, so **jet aircraft** like to fly in its lower part. Rare **nacreous (mother-of-pearl) clouds** form near the poles.
Mesosphere	About **50–80/85 km**. Temperature falls again to about **−90°C**, the **coldest** part of the atmosphere. **Meteors burn up** here.
Thermosphere and ionosphere	Above about **85 km**; very thin air, but temperature **rises sharply** as gases absorb solar energy. The **ionosphere** (about **80–400 km**) has **electrically charged particles (ions)** that **reflect radio waves** back to Earth (long-distance radio), and is where **auroras** glow near the poles. The **International Space Station** orbits here (~400 km).
Exosphere	The outermost layer, slowly merging into space; very light gases (**hydrogen, helium**).

3. How the Earth is heated

- **Insolation** (**in**coming **sol**ar radi**ation**) arrives as **short waves**. It is greatest where the Sun's rays are **vertical** (tropics) and least where they are **slanting** (poles); it also depends on **length of day**, **cloud cover** and **dust**. **Albedo** is the share of sunlight **reflected**: Earth's average is about **30%**; **fresh snow** reflects up to **80–90%**, forests and oceans much less.
- **Heat budget:** of **100 units** of sunlight reaching the top of the atmosphere, about **35** are reflected back (albedo), about **14** are absorbed by the atmosphere and about **51** by the Earth's surface. The Earth sends all this back as **long-wave terrestrial radiation**, so its average temperature stays steady. Because the atmosphere is **heated mainly from below** by the ground, it is warmest near the surface.
- **Ways air is heated: conduction** (heat passing by **contact** with warm ground), **convection** (warm air **rises vertically**, forming clouds), **advection** (**horizontal** movement of heat by wind, e.g. the hot **'loo'** winds of north India), and **radiation**. **Temperature inversion:** sometimes air near the ground is **colder** than the air above, e.g. in **valleys on clear, calm winter nights** when cold, heavy air drains down slopes; it traps **fog and smog** (as in Delhi's winters). **Isotherms** join places of equal temperature; the **thermal equator** (the belt of highest temperature) lies a little **north** of the equator because the north has more land.

4. Air pressure and pressure belts

- **Air pressure** is the **weight of the air column** above a place. At **mean sea level** it is about **1013.25 millibars (hPa)**. It **falls with height** (about **1 mb for every 10 m** near the ground), which is why mountaineers feel breathless and food takes longer to cook. It is measured with a **barometer** (**mercury** or **aneroid**); a **barograph** records it. Lines joining places of equal pressure are **isobars**. **Warm air rises → low pressure**; **cool, sinking air → high pressure**. Winds blow **from high to low pressure**.
- **Seven pressure belts:** the **Equatorial Low** (**0°–5°**; calm 'doldrums', the **ITCZ**, caused by **heat**); two **Sub-tropical Highs** (about **30°**; calm '**horse latitudes**', caused by **sinking air**; most hot deserts lie here); two **Sub-polar Lows** (about **60°**, caused by the **Earth's rotation**); and two **Polar Highs** (caused by **cold**). The belts **shift north in June and south in December** with the overhead Sun, which helps explain India's **monsoon**.

Pressure belts (red = low, blue = high)

EXAM TRAP

N_2 ~78%, O_2 ~21%, Ar ~0.93%, CO_2 ~0.04% (420+ ppm); water vapour **0–4%**; dust = **hygroscopic nuclei**; ozone **15–35 km**; **99% of air within ~32 km**; troposphere **8 km (poles) / 18 km (equator)**, all weather, **6.5°C/km**; **tropopause**; stratosphere to **~50 km**, ozone, temp **rises**, jets; mesosphere to **~85 km**, **coldest (−90°C)**, meteors; thermosphere, **ionosphere (80–400 km)** reflects radio, auroras; exosphere; insolation short-wave; **albedo ~30%** (snow 80–90%); heat budget **35 / 14 / 51**; terrestrial radiation long-wave; **conduction, convection (vertical), advection (horizontal, loo), radiation**; **inversion** (valleys, winter nights, fog, smog); isotherms; thermal equator **north** of equator; **MSL 1013.25 mb**; **1 mb per 10 m**; barometer (mercury, aneroid); isobars; belts: **equatorial low (doldrums, ITCZ), sub-tropical high 30° (horse latitudes, deserts), sub-polar low 60°, polar high**; shift with Sun.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Nitrogen is the most abundant gas in the atmosphere.

2. The ozone layer lies mainly in the stratosphere.

3. Temperature in the troposphere generally rises with height.

4. The ionosphere reflects radio waves back to the Earth.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. In the troposphere temperature falls with height, by about 6.5°C per km.

PRACTICE

Match List-I (process) with List-II (description):

List-I (process)	List-II (description)
A. Conduction	1. Long-wave heat given off by the Earth's surface
B. Convection	2. Horizontal transfer of heat by wind, such as the loo
C. Advection	3. Vertical rise of heated air
D. Terrestrial radiation	4. Transfer of heat by direct contact

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Conduction: contact; convection: vertical; advection: horizontal; terrestrial radiation: long-wave.

PRACTICE

Which one of the following statements is NOT correct?

A. The mesosphere is the coldest layer of the atmosphere.

B. The normal lapse rate is about 6.5°C per km.

C. Mean sea-level pressure is about 1013 millibars.

D. The troposphere is thickest over the poles.

Answer: D. The troposphere is thickest (about 18 km) over the equator, where strong heating lifts the air.

PRACTICE

Assertion (A): Temperature inversion is common in mountain valleys on clear winter nights.
Reason (R): Cold, dense air drains down the slopes and collects on the valley floor below warmer air.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Cold air pooling below warm air is exactly what creates the inversion.

PRACTICE

Long-distance jet airliners often cruise in the lower stratosphere. What is the main reason?

A. It contains most of the water vapour and clouds

B. It is almost free of clouds, storms and turbulence

C. It is the hottest layer of the atmosphere

D. It has the highest air pressure

Answer: B. Weather stays in the troposphere; the calm, clear stratosphere gives smoother flights, and thinner air also reduces drag.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the gases with percentages and the role of water vapour, dust and ozone. 2. Write the five layers with heights, temperature trend and one fact each. 3. Write insolation, albedo, heat budget, heating processes, inversion, pressure facts and the seven belts.

ECONOMY · Occupational pattern and sectoral distribution of income

UNDERSTAND IT FIRST

What is it? The **occupational pattern** shows **what work people do**: how many are in **farming**, in **industry** or in **services**. The **sectoral distribution of income** shows **how much each sector contributes** to national income (GVA). In rich countries most people work in **services**; in India, **nearly half of workers are still in agriculture**, but agriculture produces **less than a fifth** of output. This gap explains **low rural incomes, disguised unemployment and the need for jobs** in industry and services. Learn it as **three sectors → theories → India's data and trends → labour-force terms → types of unemployment → informal sector and reforms**.

1. Sectors and theories

Sector	Activities
Primary	Using natural resources directly: **agriculture, animal husbandry, forestry, fishing, mining and quarrying**.
Secondary (industry)	Turning raw materials into goods: **manufacturing**, **construction**, and **electricity, gas and water supply**.
Tertiary (services)	**Trade, hotels, transport, communication, banking, insurance, real estate, IT, public administration, education, health**. Some writers call knowledge work (research, IT) **quaternary**.

- **Clark–Fisher hypothesis** (**Colin Clark** and **Allan Fisher**, 1930s–40s): as an economy develops, workers and income **shift from primary to secondary and then to tertiary** activities. **Arthur Lewis's dual-economy model (1954)**: a backward farm sector with **surplus labour** supplies workers to a modern industrial sector at low wages until the surplus is used up. **India's path is different**: output moved straight from agriculture to **services** ('**service-led growth**'), while **manufacturing stayed small** (about **17%** of GVA) and too few workers left farming.

2. India: output vs jobs

Share in output vs share in jobs

- **Output (GVA, approx. 2023–24): agriculture and allied ~18%**, **industry ~28%** (manufacturing ~17%), **services ~54%**. In the **1950s** agriculture gave over **half** of national income. **Jobs (PLFS 2023–24): agriculture ~46%**, industry ~25% (construction ~12%, manufacturing ~11%), services ~29%. Agriculture's

share of jobs fell from about **70%** in 1951 to about **42%** in 2019–20, then **rose** after COVID as workers returned to villages.

- **What it means:** output per worker in farming is very low, so **many farm families are poor**; much farm work hides **disguised unemployment**. **Construction** has become the main exit from farming for unskilled workers. **AP** has a **larger farm-and-fisheries share** of GSVA than India as a whole (strong in **aquaculture, horticulture and rice**), with services next.

3. Measuring work and unemployment

Term	Meaning
PLFS	The **Periodic Labour Force Survey** by the **NSO** (since **2017**) gives annual and quarterly figures; from **2025** it also gives **monthly** estimates. (It replaced the NSSO's five-yearly surveys.)
LFPR, WPR, UR	**Labour Force Participation Rate** = (employed + unemployed seeking work) ÷ population; **Worker Population Ratio** = employed ÷ population; **Unemployment Rate** = unemployed ÷ **labour force**. For ages 15+ in 2023–24 (usual status): LFPR about **60%**, female LFPR about **42%**, UR about **3%**.
Usual status vs CWS	**Usual status** looks at the main activity over the **last 365 days** (principal + subsidiary); **Current Weekly Status (CWS)** looks at the **last 7 days**, so it catches short-term unemployment.
Types of unemployment	**Disguised** (more people on a farm than needed; removing some does not reduce output; **marginal productivity zero**); **seasonal** (idle between seasons); **structural** (skills do not match jobs as the economy changes); **frictional** (short gap while changing jobs); **cyclical** (during recessions); **educated** unemployment.

- **Informal work:** about **90%** of India's workers are in the **informal sector**, with **no written contract, paid leave or social security**. The **e-Shram portal (August 2021)** registers unorganised workers (over **30 crore**). The **four Labour Codes** (**Wages 2019; Industrial Relations, Social Security, and Occupational Safety, Health and Working Conditions 2020**) merged **29 central labour laws** and were **brought into force in November 2025**.
- **Demographic dividend:** about **two-thirds** of Indians are of working age (**15–64**) and the median age is under **30**. This is a growth chance only if people get **skills, health and productive jobs**; low **female participation** is a big gap.

EXAM TRAP

Primary (agri, forestry, fishing, **mining**); secondary (**manufacturing, construction, electricity–gas–water**); tertiary (trade, transport, finance, IT, admin); **Clark–Fisher** (primary → secondary → tertiary); **Lewis 1954** (surplus labour, dual economy); India: **service-led**, manufacturing **~17%**; GVA **~18 / 28 / 54**; jobs **~46 / 25 / 29** (PLFS 2023–24); agri jobs **~70% (1951) → ~42% (2019–20) → ~46%**; **PLFS 2017 (NSO)**, monthly from **2025**; **LFPR** (employed + unemployed ÷ population), **WPR**, **UR** (unemployed ÷ labour force); LFPR **~60%**, female **~42%**, UR **~3%**; **usual status (365 days)** vs **CWS (7 days)**; **disguised (MP = 0)**, seasonal, structural,

frictional, cyclical, educated; **~90% informal**; **e-Shram 2021 (30 crore+)**; **4 Labour Codes (29 laws), in force Nov 2025**; working age **~2/3**; AP: larger farm-and-fisheries share.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about India:

1. Agriculture employs the largest share of the workforce.

2. Services contribute the largest share of GVA.

3. Construction is part of the secondary sector.

4. Mining and quarrying are part of the tertiary sector.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Mining and quarrying belong to the primary sector because they extract natural resources.

PRACTICE

Match List-I (type of unemployment) with List-II (example):

List-I (type of unemployment)	List-II (example)
A. Disguised	1. A worker between leaving one job and joining another
B. Seasonal	2. Typists losing work as offices switch to computers
C. Structural	3. Farm labourers idle between harvest and the next sowing
D. Frictional	4. Six family members doing work that three could do on a small farm

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) Disguised: surplus family labour; seasonal: between crops; structural: technology change; frictional: between jobs.

PRACTICE

Which one of the following statements is NOT correct?

A. The Periodic Labour Force Survey is conducted by the National Statistical Office.

B. The unemployment rate is the number of unemployed divided by the labour force.

C. The Clark–Fisher hypothesis says workers move from services to agriculture as an economy develops.

D. The labour force includes both the employed and the unemployed who are seeking work.

Answer: C. The hypothesis says the shift is from primary to secondary to tertiary, not the reverse.

PRACTICE

Assertion (A): India's services sector grew rapidly after the 1991 reforms.
Reason (R): The Clark–Fisher hypothesis was put forward by Colin Clark and Allan Fisher.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but who proposed the hypothesis does not explain why India's services grew (IT, telecom, finance and liberalisation did).

PRACTICE

Five brothers work together on a two-acre family farm. Two of them take jobs in the city, and the farm's output stays the same. What kind of unemployment did the two brothers have earlier?

A. Open unemployment

B. Frictional unemployment

C. Disguised unemployment

D. Cyclical unemployment

Answer: C. They seemed employed, but their marginal productivity was zero: classic disguised unemployment.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the three sectors with examples and the Clark–Fisher and Lewis ideas. 2. Write India's output and job shares, the trend and what they mean. 3. Write PLFS terms, types of unemployment, informal work, e-Shram, the Labour Codes and the demographic dividend.

SCIENCE · Indian wildlife and forest laws, projects and recent initiatives

UNDERSTAND IT FIRST

What is it? Day 27 covered the **Wildlife Protection Act, the IUCN list, CITES and biosphere reserves**. Today completes the picture: the **other laws** that protect **forests** and forest people, the **institutions** that study and guard wildlife, the **species projects** in more detail, and the **new schemes since 2020** that exams love to ask about. Learn it as **laws in time order → institutions → species projects → recent initiatives → forest cover → AP**.

1. Laws in time order

Law	Key points
Indian Forest Act, 1927	Colonial law classifying forests as **reserved, protected and village forests**; the **2017 amendment** said **bamboo grown outside forests is not a 'tree'**, so farmers can cut and sell it freely.
Prevention of Cruelty to Animals Act, 1960	Punishes cruelty to animals; set up the **Animal Welfare Board of India**.
Wildlife (Protection) Act, 1972	Hunting ban, protected areas, NTCA, WCCB, four schedules (see Day 27).
Forest (Conservation) Act, 1980	Forest land cannot be **diverted** for non-forest use (mining, dams, roads) **without the Union Government's prior approval**. In **T.N. Godavarman (1996)** the Supreme Court said 'forest' means **any land that is forest in the dictionary sense**, whatever its record. The **2023 amendment** renamed it the **Van (Sanrakshan Evam Samvardhan) Adhiniyam**, limited it mainly to **recorded forests**, and exempted some **strategic projects near borders**; the Supreme Court has asked that the broad meaning of 'forest' be kept while cases are heard.

Environment (Protection) Act, 1986	An **umbrella law** passed after the **Bhopal gas tragedy (1984)**; the basis for Environmental Impact Assessment and coastal rules.
Forest Rights Act, 2006	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act: recognises the rights of **forest-dwelling STs** and **other traditional forest dwellers** (who must show **75 years / three generations** of residence) who lived there before **13 December 2005**. **Individual rights** to land under cultivation (up to **4 hectares**) and **community forest resource rights**. The **Gram Sabha** starts the process. The nodal ministry is **Tribal Affairs**.
Compensatory Afforestation Fund Act, 2016	When forest land is diverted, the user pays for **compensatory afforestation**; the money goes to **CAMPA** funds (**90% to States**, 10% to the Centre).

2. Institutions

Botanical Survey of India (1890) and **Zoological Survey of India (1916)**, both in **Kolkata**; **Wildlife Institute of India (1982, Dehradun)**; **Salim Ali Centre for Ornithology and Natural History (SACON, Coimbatore)**; **National Green Tribunal (2010)** for environmental cases; NTCA and WCCB (Day 27). **Wildlife Week** is held in the **first week of October**.

3. Species projects and recent initiatives

Recent wildlife and forest initiatives

Project	Facts
Big cats	**Project Tiger** marked **50 years** in **2023**. **Project Lion (2020)** protects the **Asiatic lion** of **Gir (Gujarat)**, whose count was about **891 in 2025**. **Project Cheetah**: **8 cheetahs from Namibia** were released at **Kuno (MP)** on **17 September 2022**, and **12 from South Africa** came in **February 2023**; **Gandhi Sagar (MP)** became a second site in **2025**. The **International Big Cat Alliance**, launched by India in **2023**, covers **seven big cats** (tiger, lion, leopard, snow leopard, cheetah, jaguar, puma).
Other species	**Project Elephant (1992)**, with **33 elephant reserves**; **Project Snow Leopard (2009)**, with the first national count (**2024**) at about **718**; **Project Dolphin (2020)**; the **Vulture Action Plan 2020–25**; **Great Indian Bustard**: the Supreme Court (**2021**, revised **2024**) ordered steps such as **burying or marking power lines** in its habitat, plus **conservation breeding** in Rajasthan; **Olive Ridley turtles** nest in huge groups (**arribada**) at **Gahirmatha and Rushikulya** (Odisha) and along the AP coast; the Coast Guard runs **'Operation Olivia'** to protect them.
Recent schemes	**National Wildlife Action Plan (2017–2031)**; **Nagar Van Yojana (2020)**: urban forests; **MISHTI (2023)**: **mangrove** planting along the coast; **Amrit Dharohar (2023)**: care of **Ramsar wetlands**; **Green Credit Programme (2023)**: tradable credits for actions like tree planting; **'Ek Ped Maa Ke Naam' (2024)**: tree planting in mothers' names; **Green India Mission** (one of the NAPCC missions).

- **Forest cover (India State of Forest Report 2023):** forest cover about **21.8%** of India's area, and **forest plus tree cover about 25.2%**; the **National Forest Policy (1988)** goal is **33%**. The report is published every two years by the **Forest Survey of India (Dehradun)**.
- **AP: Nagarjunasagar–Srisailam Tiger Reserve** (India's largest); the **Red Sanders Anti-Smuggling Task Force** guards **Seshachalam**; **Coringa and Krishna** mangroves benefit from **MISHTI**; **Olive Ridley** nesting beaches from **Srikakulam to Nellore**; **Rollapadu** for the Great Indian Bustard; **Kolleru** under Amrit Dharohar.

EXAM TRAP

Indian Forest Act 1927 (reserved, protected, village; **2017 bamboo**); **PCA 1960** (AWBI); **WPA 1972**; **FCA 1980** (Centre's prior approval), **Godavarman 1996** (dictionary meaning), **2023 → Van (Sanrakshan Evam Samvardhan) Adhiniyam**; **EPA 1986** (after Bhopal); **FRA 2006** (Tribal Affairs, **13 Dec 2005**, OTFD **75 years**, **4 ha**, CFR, **Gram Sabha**); **CAMPA Act 2016 (90:10)**; **BSI 1890, ZSI 1916** (Kolkata), **WII 1982** (Dehradun), **SACON**, **NGT 2010**; Wildlife Week **first week of October**; **Project Tiger 50 (2023)**; **Project Lion 2020 (Gir ~891, 2025)**; **Cheetah: Namibia 8 (17 Sept 2022), SA 12 (Feb 2023), Gandhi Sagar 2025**; **IBCA 2023 (7 cats)**; **Elephant 1992 (33 reserves)**; **Snow Leopard 2009 (~718)**; **Dolphin 2020**; **Vulture plan 2020–25**; **GIB power lines (SC 2021/2024)**; **Olive Ridley arribada, Operation Olivia**; **NWAP 2017–31**; **Nagar Van 2020**; **MISHTI, Amrit Dharohar, Green Credit 2023**; **Ek Ped Maa Ke Naam 2024**; **ISFR 2023: 21.8% forest, 25.2% total; target 33%**; AP: **NSTR, RSASTF, Coringa, Rollapadu, Kolleru**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Forest Rights Act, 2006:
1. It recognises the rights of forest-dwelling Scheduled Tribes and other traditional forest dwellers.
2. The Gram Sabha starts the process of deciding rights.
3. Individual rights to land under cultivation can extend up to 4 hectares.
4. The Ministry of Environment, Forest and Climate Change is its nodal ministry.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The nodal ministry for the Forest Rights Act is the Ministry of Tribal Affairs.

PRACTICE

Match List-I (initiative) with List-II (focus):

List-I (initiative)	List-II (focus)
A. MISHTI	1. Tradable credits for environment-positive actions
B. Amrit Dharohar	2. Urban forests
C. Nagar Van Yojana	3. Ramsar wetlands
D. Green Credit Programme	4. Mangrove planting along the coast

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) MISHTI: mangroves; Amrit Dharohar: wetlands; Nagar Van: urban forests; Green Credit: credits.

PRACTICE

Which one of the following statements is NOT correct?

A. The 2023 amendment renamed the Forest (Conservation) Act as the Van (Sanrakshan Evam Samvardhan) Adhiniyam.

B. CAMPA funds are used for compensatory afforestation.

C. The first cheetahs of Project Cheetah came from Namibia.

D. Project Lion protects the Asiatic lions of Assam.

Answer: D. Asiatic lions live in and around Gir in Gujarat.

PRACTICE

Assertion (A): Olive Ridley turtles nest in mass arrivals called arribadas at Gahirmatha and Rushikulya in Odisha.
Reason (R): Olive Ridley turtles are protected under Schedule I of the Wildlife (Protection) Act.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but legal protection does not explain why the turtles nest in mass groups; that is their natural behaviour.

PRACTICE

A company wants to use 50 hectares of recorded forest land in AP for a mining project. What does the law require?

A. Only the State Government's permission

B. Approval from the National Biodiversity Authority only

C. No approval, because mining is a national need

D. The Union Government's prior approval under the forest conservation law, plus payment for compensatory afforestation (CAMPA)

Answer: D. Diverting forest land needs the Centre's prior approval; the user also pays for compensatory afforestation through CAMPA (and Forest Rights Act claims must be settled).

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the laws in time order with one key point each, including the Godavarman case and the 2023 change. 2. Write the institutions and the species projects with facts. 3. Write the recent schemes with years, forest cover figures and AP's initiatives.

NOTEBOOK PAGE · Day 28: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Atmosphere, heat and pressure

Area	Write
Composition and layers	Gases with %; water vapour, dust, ozone; five layers with heights and one fact each; boundaries
Heat and pressure	Insolation factors; albedo; heat budget; four ways of heating; lapse rate; inversion; greenhouse effect; MSL pressure; instruments; seven pressure belts

ECONOMY: Occupations and sectors

- Three sectors with examples; Clark–Fisher and Lewis; India's output vs job shares and trends; PLFS terms (LFPR, WPR, UR, usual status, CWS); types of unemployment; informal sector, e-Shram, labour codes; demographic dividend; AP's pattern.

SCIENCE: Wildlife and forest laws, projects, initiatives

Write: laws with years (1927, 1960, 1972, 1980/2023, 1986, 2002, 2006 FRA, 2016 CAMPA); institutions; species projects with facts; recent initiatives with years; forest cover figures; AP's initiatives.